

DDoS Incident Report

Report Period	2026-08-24 - 2026-08-31
Generated At	2026-08-31 18:48:58
Classification	ML: Isolation Forest + Random Forest

1. Executive Summary

Metric	Value	Attack Vector	Count
Report Period	2026-08-24 - 2026-08-31	ICMP Flood	16
Total Threats Mitigated	71	SYN Flood	21
High Priority	17	UDP Flood	19
Low Priority	54	Uncertain	15
Manual Releases	0		
Manual Blocks	0	Action	Count
Total Flows Observed	81808255772	Quarantined	10
True Negatives Passed	38974845	Time Ban	10
False Positives	298	Blackhole	10
FP Rate	0.27%	Released	9

2. Performance Benchmark

2a. Isolation Forest-Anomaly Detection

Metric	Value	Description
Precision	99.66%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	93.62%	Share of actual attacks successfully flagged
F1-Score	96.55%	Balanced measure combining Precision and Recall
Accuracy	96.94%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.27%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	6.38%	Actual attacks that went undetected
True Positive Rate (TPR)	93.62%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.73%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 87236	FN 5945
Actual: Normal	FP 298	TN 110345

2b. Random Forest-Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	38268	0	0
Act: ICMP	0	23125	0
Act: UDP	0	0	25843

2c. Response Latency

Metric	Value	Description
Detection Time	630.1 ms	Average time taken to detect an attack
Mitigation Response Time	4.8 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	0.91%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	3.70%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	3.59%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

No offence history for this period.

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-08-31 17:57:56	10.0.0.10	DDoS	Uncertain	68.7%	Low	Released
2026-08-31 18:00:25	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-08-31 18:00:25	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-08-31 18:00:26	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-08-31 18:00:30	10.0.0.17	DDoS	SYN Flood	90.8%	Critical	Time Ban (1m)
2026-08-31 18:00:31	10.0.0.17	DDoS	SYN Flood	90.8%	Critical	Time Ban
2026-08-31 18:00:31	10.0.0.16	DDoS	SYN Flood	91.0%	Critical	Time Ban (1m)
2026-08-31 18:00:31	10.0.0.18	DDoS	SYN Flood	93.3%	Critical	Time Ban (1m)
2026-08-31 18:00:31	10.0.0.16	DDoS	SYN Flood	92.2%	Critical	Time Ban
2026-08-31 18:00:31	10.0.0.18	DDoS	SYN Flood	93.3%	Critical	Time Ban
2026-08-31 18:00:33	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-08-31 18:00:38	10.0.0.19	DDoS	SYN Flood	93.3%	Critical	Time Ban (1m)
2026-08-31 18:00:39	10.0.0.19	DDoS	SYN Flood	93.3%	Critical	Time Ban
2026-08-31 18:01:30	10.0.0.17	DDoS	SYN Flood	90.8%	Critical	Time Ban (2m)
2026-08-31 18:01:31	10.0.0.16	DDoS	SYN Flood	93.9%	Critical	Time Ban (2m)
2026-08-31 18:01:31	10.0.0.18	DDoS	SYN Flood	93.8%	Critical	Time Ban (2m)
2026-08-31 18:01:38	10.0.0.19	DDoS	SYN Flood	94.2%	Critical	Time Ban (2m)
2026-08-31 18:03:30	10.0.0.17	DDoS	SYN Flood	94.2%	Critical	Released
2026-08-31 18:03:31	10.0.0.16	DDoS	SYN Flood	93.9%	Critical	Time Ban (5m)
2026-08-31 18:03:31	10.0.0.18	DDoS	SYN Flood	93.8%	Critical	Time Ban (5m)
2026-08-31 18:03:36	10.0.0.17	DDoS	SYN Flood	94.2%	Critical	Time Ban (5m)
2026-08-31 18:03:38	10.0.0.19	DDoS	SYN Flood	94.2%	Critical	Time Ban (5m)
2026-08-31 18:07:58	10.0.0.4	DDoS	Uncertain	58.9%	Low	Released
2026-08-31 18:10:25	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-08-31 18:10:26	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-08-31 18:10:28	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-08-31 18:10:34	10.0.0.25	DDoS	ICMP Flood	91.7%	Critical	Time Ban (1m)
2026-08-31 18:10:34	10.0.0.24	DDoS	ICMP Flood	91.9%	Critical	Time Ban (1m)
2026-08-31 18:10:34	10.0.0.25	DDoS	ICMP Flood	91.7%	Critical	Time Ban
2026-08-31 18:10:34	10.0.0.24	DDoS	ICMP Flood	91.9%	Critical	Time Ban
2026-08-31 18:10:38	10.0.0.23	DDoS	ICMP Flood	91.9%	Critical	Time Ban (1m)
2026-08-31 18:10:38	10.0.0.23	DDoS	ICMP Flood	91.9%	Critical	Time Ban
2026-08-31 18:11:34	10.0.0.25	DDoS	ICMP Flood	92.1%	High	Time Ban (2m)
2026-08-31 18:11:34	10.0.0.24	DDoS	ICMP Flood	92.1%	High	Time Ban (2m)
2026-08-31 18:11:38	10.0.0.23	DDoS	ICMP Flood	92.3%	High	Time Ban (2m)
2026-08-31 18:13:34	10.0.0.25	DDoS	ICMP Flood	96.0%	High	Time Ban (5m)
2026-08-31 18:13:34	10.0.0.24	DDoS	ICMP Flood	96.0%	High	Time Ban (5m)

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-08-31 18:13:38	10.0.0.23	DDoS	ICMP Flood	96.0%	Critical	Time Ban (5m)
2026-08-31 18:20:25	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-08-31 18:20:27	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-08-31 18:20:28	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-08-31 18:20:46	10.0.0.20	DDoS	UDP Flood	76.6%	High	Time Ban (1m)
2026-08-31 18:20:46	10.0.0.20	DDoS	UDP Flood	76.6%	High	Time Ban
2026-08-31 18:20:48	10.0.0.22	DDoS	UDP Flood	86.8%	High	Time Ban (1m)
2026-08-31 18:20:49	10.0.0.21	DDoS	UDP Flood	78.8%	High	Time Ban (1m)
2026-08-31 18:20:49	10.0.0.22	DDoS	UDP Flood	86.8%	Critical	Time Ban
2026-08-31 18:20:49	10.0.0.21	DDoS	UDP Flood	78.8%	High	Time Ban
2026-08-31 18:21:46	10.0.0.20	DDoS	UDP Flood	79.0%	High	Released
2026-08-31 18:21:48	10.0.0.22	DDoS	UDP Flood	86.8%	High	Time Ban (2m)
2026-08-31 18:21:49	10.0.0.21	DDoS	UDP Flood	78.8%	High	Released
2026-08-31 18:21:52	10.0.0.20	DDoS	UDP Flood	79.0%	Critical	Time Ban (2m)
2026-08-31 18:22:10	10.0.0.21	DDoS	UDP Flood	78.8%	High	Time Ban (2m)
2026-08-31 18:23:48	10.0.0.22	DDoS	UDP Flood	86.8%	High	Time Ban (5m)
2026-08-31 18:23:52	10.0.0.20	DDoS	UDP Flood	80.7%	High	Time Ban (5m)
2026-08-31 18:24:10	10.0.0.21	DDoS	UDP Flood	78.8%	High	Time Ban (5m)
2026-08-31 18:30:54	10.0.0.24	DDoS	Uncertain	-	Low	Released
2026-08-31 18:31:05	10.0.0.24	DDoS	ICMP Flood	91.8%	Critical	Time Ban (10m)
2026-08-31 18:31:05	10.0.0.21	DDoS	UDP Flood	79.3%	Critical	Time Ban (10m)
2026-08-31 18:31:31	10.0.0.17	DDoS	SYN Flood	93.6%	Critical	Blackhole
2026-08-31 18:33:31	10.0.0.18	DDoS	SYN Flood	93.3%	Critical	Blackhole
2026-08-31 18:33:33	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Blackhole
2026-08-31 18:33:33	10.0.0.19	DDoS	SYN Flood	93.0%	Critical	Blackhole
2026-08-31 18:33:46	10.0.0.25	DDoS	ICMP Flood	96.8%	Critical	Blackhole
2026-08-31 18:34:03	10.0.0.23	DDoS	ICMP Flood	96.3%	Critical	Blackhole
2026-08-31 18:42:58	10.0.0.25	DDoS	Uncertain	-	Low	Released
2026-08-31 18:43:38	10.0.0.22	DDoS	UDP Flood	86.3%	Critical	Released
2026-08-31 18:43:38	10.0.0.22	DDoS	UDP Flood	86.3%	Critical	Blackhole
2026-08-31 18:43:39	10.0.0.20	DDoS	UDP Flood	79.7%	Critical	Blackhole
2026-08-31 18:44:04	10.0.0.24	DDoS	ICMP Flood	91.6%	Critical	Blackhole
2026-08-31 18:45:49	10.0.0.21	DDoS	UDP Flood	85.9%	Critical	Blackhole
2026-08-31 18:48:04	10.0.0.3	DDoS	Uncertain	58.9%	Low	Released

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE